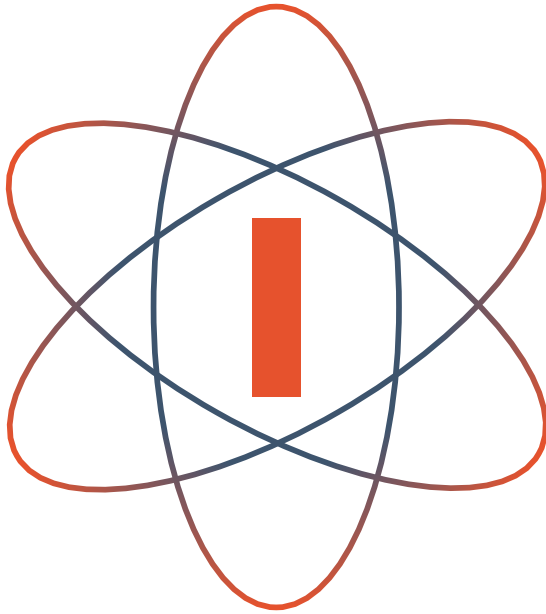




IN THE 1950s, the father of the country's nuclear programme, Dr. Homi Bhabha, conceived an ambitious three-stage nuclear programme. This happened long before India had mapped its uranium deposits and had no technology to utilise thorium. The plan: use natural uranium in pressurised heavy water reactors (PHWRs), then plutonium in fast breeder reactors (FBRs), and finally, thorium-based reactors to achieve a self-sustaining nuclear fuel cycle. The endgame was clear: each stage would feed the next, culminating in a self-sustaining thorium-uranium fuel cycle to power the country and cut down crude oil imports.

For decades, this vision remained a promise. But in April 2026, India crossed a crucial milestone. At Kalpakkam in Tamil Nadu, the prototype FBR achieved criticality, marking the country's entry into the second stage of Bhabha's blueprint. This is no small achievement for India, which is the second nation after Russia to accomplish this, albeit with delays. India's present nuclear capacity of 8.8 GW is around 2% of its energy mix. The target is 22GW by 2032 and 100 GW by 2047.

With energy security under the spotlight due to the West Asia war and clean energy being the focal point of discussions about building a sustainable future, the penultimate stage of commercial deployment of thorium-based reactors is important (India has abundant thorium reserves).

However, owing to technological complexities, achieving that could take decades. Another hiccup is availability of uranium. Even though recent discoveries have increased supplies, they are not enough. India imports 70% of its annual 1,800-2,000-tonne requirement. Major suppliers are Kazakhstan (35-40%), Canada (30-35%), with a small share coming from Namibia, Australia, Uzbekistan, and Russia.

In such a scenario, does India need to fast-track the three-stage programme to reach the thorium stage? Or can it come up with new technologies that can reduce the time taken to become *atmanirbhar* in the nuclear fuel cycle? This includes introducing thorium in Stage 1 itself, rather than waiting for Stage 3 or fusion technology that does not require uranium or plutonium as fuel, thus completely skipping Stage 2.

NEW VISTAS

Some experts, like the former chairman of the Atomic Energy

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ENERGY MIX**



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